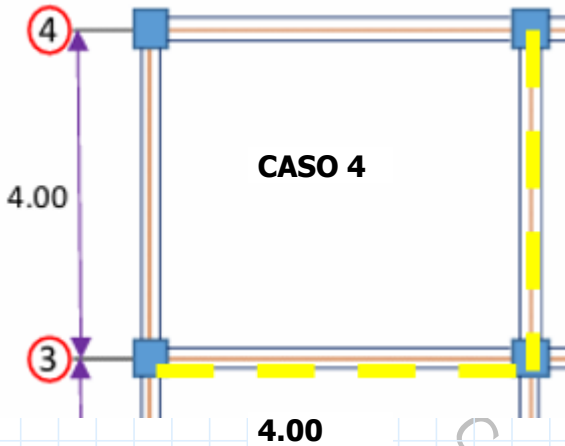


$$h_{losa} := 0.14 \text{ m}$$

NIVEL 1 - PAÑO VII

3.- CÁLCULO DE COEFICIENTE



$$La := 4.00 \text{ m}$$

$$Lb := 4.00 \text{ m}$$

$$m := \frac{La}{Lb} = 1$$

3.1. MOMENTO NEGATIVO (CV+CM) : M(-) CV + CM:

INTERPOLAR

$$x_{1a} := 1.00 \quad y_{1a} := 0.050 \text{ m} = 1$$

$$x_{2a} := 0.00 \quad y_{2a} := 0.050$$

$$Ca_{neg} := \left(\frac{m - x_{1a}}{x_{2a} - x_{1a}} \right) (y_{2a} - y_{1a}) + y_{1a} = 0.05$$

$$x_{1b} := 1.00 \quad y_{1b} := 0.050$$

$$Cb_{neg} := \left(\frac{m - x_{1b}}{x_{2b} - x_{1b}} \right) (y_{2b} - y_{1b}) + y_{1b} = 0.05$$

Relacion		Caso 4
m=La/Lb		
1	Ca - neg	0.050
	Cb - neg	0.050

3.2. MOMENTO POSITIVO (CM) : M(+) CM:

INTERPOLAR

$$x_{1aP} := 1.00 \quad y_{1aP} := 0.027 \quad m = 1$$

$$x_{2aP} := 0.00 \quad y_{2aP} := 0.027$$

$$Ca_{DLP} := \left(\frac{m - x_{1aP}}{x_{2aP} - x_{1aP}} \right) (y_{2aP} - y_{1aP}) + y_{1aP} = 0.027$$

$$x_{1bP} := 1.00 \quad y_{1bP} := 0.027$$

$$x_{2bP} := 0.00 \quad y_{2bP} := 0.027$$

$$Cb_{DLP} := \left(\frac{m - x_{1bP}}{x_{2bP} - x_{1bP}} \right) (y_{2bP} - y_{1bP}) + y_{1bP} = 0.027$$

Relacion		Caso 4
m=La/Lb		
1	Ca - DL	0.027
	Cb - DL	0.027
0.95	Ca - DL	0.030
	Cb - DL	0.024



3.3. MOMENTO POSITIVO (CV) : M(+) CV:

INTERPOLAR

$$x_{1aL} := 1.00 \quad y_{1aL} := 0.032 \quad m = 1$$

$$x_{2aL} := 0.00 \quad y_{2aL} := 0.032$$

$$Ca_{LLP} := \left(\frac{m - x_{1aL}}{x_{2aL} - x_{1aL}} \right) (y_{2aL} - y_{1aL}) + y_{1aL} = 0.032$$

$$x_{1bL} := 1.00 \quad y_{1bL} := 0.032$$

$$x_{2bL} := 0.00 \quad y_{2bL} := 0.032$$

$$Cb_{LLP} := \left(\frac{m - x_{1bL}}{x_{2bL} - x_{1bL}} \right) (y_{2bL} - y_{1bL}) + y_{1bL} = 0.032$$

Relacion		Caso 4
m=La/Lb		
1	Ca-DL	0.032
	Cb-DL	0.032
0.95	Ca-DL	0.035
	Cb-DL	0.029

4.- CÁLCULO DE CARGAS

DATOS DE DISEÑO

$$f'c := 210 \frac{kg}{cm^2}$$

$$f'y := 4200 \frac{kg}{cm^2}$$

$$Pc := 2400 \frac{kg}{cm^3}$$

$$\gamma_c := 2.40 \frac{ton}{m^3}$$

$$Pacab := 100 \frac{kg}{m^2}$$

$$Palb := 300 \frac{kg}{m^2}$$

CM

$$CM := 0.736 \frac{ton}{m^2}$$

Descripción	L(m)	A(m)	h	Yc	Peso
Peso Propio	1.00	1.00	0.14	2.4	0.3360
Peso acab.	1.00	1.00		0.1	0.1000
Peso alb.	1.00	1.00		0.30	0.3000
					0.7360

CV

$$CV := 0.200 \frac{ton}{m^2}$$

Descripción	L(m)	A(m)	h	Yc	Peso
Peso S/C	1.00	1.00		0.2	0.200
					0.200

$$WL := 1.60 \quad CV = 0.32 \frac{ton}{m^2}$$



$$WD := 1.20 \text{ CM} = 0.883 \frac{\text{ton}}{\text{m}^2}$$

$$Wu := 1.60 \text{ CV} + 1.20 \text{ CM} = 1.203 \frac{\text{ton}}{\text{m}^2}$$

5.- CÁLCULO DE MOMENTOS

5.1. CÁLCULO DE MOMENTOS NEGATIVOS (CM+CV): M(-)WD+WL:

$$Ma_{neg} := Ca_{neg} \cdot (WD + WL) \cdot (La)^2 = 0.963 \text{ ton}$$

$$Mb_{neg} := Cb_{neg} \cdot (WD + WL) \cdot (Lb)^2 = 0.963 \text{ ton}$$

5.2. CÁLCULO DE MOMENTOS POSITIVOS (CM): M(+)WD:

$$Ma_{wd} := Ca_{DLP} \cdot (WD) \cdot (La)^2 = 0.382 \text{ ton}$$

$$Mb_{wd} := Cb_{DLP} \cdot (WD) \cdot (Lb)^2 = 0.382 \text{ ton}$$

5.3. CÁLCULO DE MOMENTOS POSITIVOS (CV): M(+) WL:

$$Ma_{wl} := Ca_{LLP} \cdot (WL) \cdot (La)^2 = 0.164 \text{ ton}$$

$$Mb_{wl} := Cb_{LLP} \cdot (WL) \cdot (Lb)^2 = 0.164 \text{ ton}$$

5.4. resumen de momentos positivos:

$$Ma_{res} := Ma_{wd} + Ma_{wl} = 0.545 \text{ ton}$$

$$Ma_{neg} = 0.963 \text{ ton}$$

$$Mb_{res} := Mb_{wd} + Mb_{wl} = 0.545 \text{ ton}$$

$$Mb_{neg} = 0.963 \text{ ton}$$

NIVEL 2 - PAÑO VII

3.- CÁLCULO DE COEFICIENTE

CM

$$CM_2 := 0.736 \frac{\text{ton}}{\text{m}^2}$$

Descripción	L(m)	A(m)	h	Yc	Peso
Peso Propio	1.00	1.00	0.14	2.4	0.3360
Peso acab.	1.00	1.00		0.1	0.1000
Peso alb.	1.00	1.00		0.30	0.3000
					0.7360



CV

$$CV_2 := 0.250 \frac{\text{ton}}{\text{m}^2}$$

Descripción	L(m)	A(m)	h	Yc	Peso
Peso S/C	1.00	1.00		0.25	0.250
					0.250

$$WL_2 := 1.60 \text{ CV}_2 = 0.4 \frac{\text{ton}}{\text{m}^2}$$

$$WD_2 := 1.20 \text{ CM}_2 = 0.883 \frac{\text{ton}}{\text{m}^2}$$

$$Wu_2 := 1.60 \text{ CV}_2 + 1.20 \text{ CM}_2 = 1.283 \frac{\text{ton}}{\text{m}^2}$$

5.- CÁLCULO DE MOMENTOS

5.1. CÁLCULO DE MOMENTOS NEGATIVOS (CM+CV): M(-)WD+WL:

$$Ma_{neg2} := Ca_{neg} \cdot (WD_2 + WL_2) \cdot (La)^2 = 1.027 \text{ ton}$$

$$Mb_{neg2} := Cb_{neg} \cdot (WD_2 + WL_2) \cdot (Lb)^2 = 1.027 \text{ ton}$$

5.2. CÁLCULO DE MOMENTOS POSITIVOS (CM): M(+)WD:

$$Ma_{wd2} := Ca_{DLP} \cdot (WD_2) \cdot (La)^2 = 0.382 \text{ ton}$$

$$Mb_{wd2} := Cb_{DLP} \cdot (WD_2) \cdot (Lb)^2 = 0.382 \text{ ton}$$

5.3. CÁLCULO DE MOMENTOS POSITIVOS (cv): M(+)WL:

$$Ma_{wl2} := Ca_{LLP} \cdot (WL_2) \cdot (La)^2 = 0.205 \text{ ton}$$

$$Mb_{wl2} := Cb_{LLP} \cdot (WL_2) \cdot (Lb)^2 = 0.205 \text{ ton}$$

5.4. resumen de momentos positivos:

$$Ma_{res2} := Ma_{wd2} + Ma_{wl2} = 0.586 \text{ ton}$$

$$Ma_{neg2} = 1.027 \text{ ton}$$

$$Mb_{res2} := Mb_{wd2} + Mb_{wl2} = 0.586 \text{ ton}$$

$$Mb_{neg2} = 1.027 \text{ ton}$$

NIVEL 3- PAÑO VII

3.- CÁLCULO DE COEFICIENTE

CM

$$CM_3 := 0.736 \frac{\text{ton}}{\text{m}^2}$$

Descripción	L(m)	A(m)	h	Yc	Peso
Peso Propio	1.00	1.00	0.14	2.4	0.3360
Peso acab.	1.00	1.00		0.1	0.1000
Peso alb.	1.00	1.00		0.30	0.3000
					0.7360

CV

$$CV_3 := 0.350 \frac{\text{ton}}{\text{m}^2}$$

Descripción	L(m)	A(m)	h	Yc	Peso
Peso S/C	1.00	1.00		0.35	0.350
					0.350

$$WL_3 := 1.60 \quad CV_3 = 0.56 \frac{\text{ton}}{\text{m}^2} \quad WD_3 := 1.20 \quad CM_3 = 0.883 \frac{\text{ton}}{\text{m}^2}$$

$$Wu_3 := 1.60 \quad CV_3 + 1.20 \quad CM_3 = 1.443 \frac{\text{ton}}{\text{m}^2}$$

5.- CÁLCULO DE MOMENTOS

5.1. CÁLCULO DE MOMENTOS NEGATIVOS (CM+CV): M(-)WD+WL:

$$Ma_{neg3} := Ca_{neg} \cdot (WD_3 + WL_3) \cdot (La)^2 = 1.155 \text{ ton}$$

$$Mb_{neg3} := Cb_{neg} \cdot (WD_3 + WL_3) \cdot (Lb)^2 = 1.155 \text{ ton}$$

5.2. CÁLCULO DE MOMENTOS POSITIVOS (CM): M(+)WD:

$$Ma_{wd3} := Ca_{DLP} \cdot (WD_3) \cdot (La)^2 = 0.382 \text{ ton}$$

$$Mb_{wd3} := Cb_{DLP} \cdot (WD_3) \cdot (Lb)^2 = 0.382 \text{ ton}$$

5.3. CÁLCULO DE MOMENTOS POSITIVOS (cv): M(+)WI:

$$Ma_{wl3} := Ca_{LLP} \cdot (WL_3) \cdot (La)^2 = 0.287 \text{ ton}$$

$$Mb_{wl3} := Cb_{LLP} \cdot (WL_3) \cdot (Lb)^2 = 0.287 \text{ ton}$$



5.4. resumen de momentos positivos:

$$Ma_{res3} := Ma_{wd3} + Ma_{wl3} = 0.668 \text{ ton}$$

$$Ma_{neg3} = 1.155 \text{ ton}$$

$$Mb_{res3} := Mb_{wd3} + Mb_{wl3} = 0.668 \text{ ton}$$

$$Mb_{neg3} = 1.155 \text{ ton}$$

NIVEL 4 - PAÑO VII

3.- CÁLCULO DE COEFICIENTE

$$CM_4 := 0.736 \frac{\text{ton}}{\text{m}^2}$$

Descripción	L(m)	A(m)	h	Yc	Peso
Peso Propio	1.00	1.00	0.14	2.4	0.3360
Peso acab.	1.00	1.00		0.1	0.1000
Peso alb.	1.00	1.00		0.30	0.3000
					0.7360

CV

$$CV_4 := 0.450 \frac{\text{ton}}{\text{m}^2}$$

Descripción	L(m)	A(m)	h	Yc	Peso
Peso S/C	1.00	1.00		0.45	0.450
					0.450

$$WL_4 := 1.60 \text{ CV}_4 = 0.72 \frac{\text{ton}}{\text{m}^2}$$

$$WD_4 := 1.20 \text{ CM}_4 = 0.883 \frac{\text{ton}}{\text{m}^2}$$

$$Wu_4 := 1.60 \text{ CV}_4 + 1.20 \text{ CM}_4 = 1.603 \frac{\text{ton}}{\text{m}^2}$$

5.- CÁLCULO DE MOMENTOS

5.1. CÁLCULO DE MOMENTOS NEGATIVOS (CM+CV): M(-)WD+WL:

$$Ma_{neg4} := Ca_{neg} \cdot (WD_4 + WL_4) \cdot (La)^2 = 1.283 \text{ ton}$$

$$Mb_{neg4} := Cb_{neg} \cdot (WD_4 + WL_4) \cdot (Lb)^2 = 1.283 \text{ ton}$$

5.2. CÁLCULO DE MOMENTOS POSITIVOS (CM): M(+)WD:

$$Ma_{wd4} := Ca_{DLP} \cdot (WD_4) \cdot (La)^2 = 0.382 \text{ ton}$$



$$Mb_{wd4} := Cb_{DLP} \cdot (WD_4) \cdot (Lb)^2 = 0.382 \text{ ton}$$

5.3. CÁLCULO DE MOMENTOS POSITIVOS (cv): M(+)WI:

$$Ma_{wl4} := Ca_{LLP} \cdot (WL_4) \cdot (La)^2 = 0.369 \text{ ton}$$

$$Mb_{wl4} := Cb_{LLP} \cdot (WL_4) \cdot (Lb)^2 = 0.369 \text{ ton}$$

5.4. resumen de momentos positivos:

$$Ma_{res4} := Ma_{wd4} + Ma_{wl4} = 0.75 \text{ ton}$$

$$Ma_{neg4} = 1.283 \text{ ton}$$

$$Mb_{res4} := Mb_{wd4} + Mb_{wl4} = 0.75 \text{ ton}$$

$$Mb_{neg4} = 1.283 \text{ ton}$$